

Hybrid Model V1 – Summary

Model Idea:

This is a model combining the friction imbalance model and scratch model already created. The force calculated each timestep from the scratch model and the friction imbalance model are combined and the force is applied on the curling stone. Below is a quick description of the two models as well as any new changes.

Friction Imbalance Model:

The running band heats up pebbles in contact with it, when the pebble is no longer in contact, the pebble cools down. This is shown in Figure 1 and explained in more depth in the summary for this model.

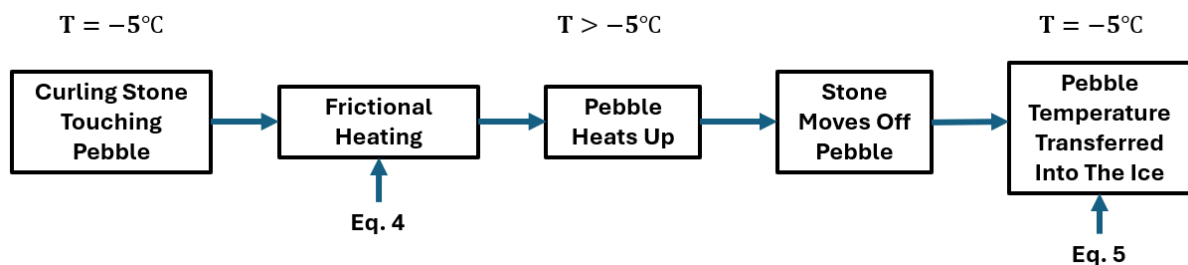


Figure 1: A diagram showing how the process of pebble heating and cooling happens, with temperatures labelled as well as the corresponding equations (that can be found in the friction imbalance summary).

Scratch Model:

Scratches are created on the surface of pebbles as after the front of the running band has passed over them, the scratch is created due to the rotation of the curling stone as it moves. Therefore, the scratches are created at an angle tangent to the running band at the point it touches the pebble. This is explained in more depth in the scratch model summary.

Model Changes From Previous Models:

- Set the constant A to 0 when the initial angular velocity is 0
- Set the model so that it runs until both x and y velocities of the curling stone are zero instead of for a fixed time period

Assumptions:

All pebbles have the same contact area and pebbles are distributed randomly across the icesheet.

All pebbles heat and cool at the same rate and all excess heat is conducted into the ice.

Constant coefficient of friction acting on the angular velocity.

Ignoring the height of pebbles.

Predictions:

Predict this model will give the same results as the scratch model dependant on initial conditions, this is because the friction imbalance model produces a small curl.

Predict the curling stone to curl in the same direction as its direction of rotation.

Predict the curling stone to have a curl greater than a few cm and less than 2m, and for its curl to be directly proportional to the constant A.

Results:

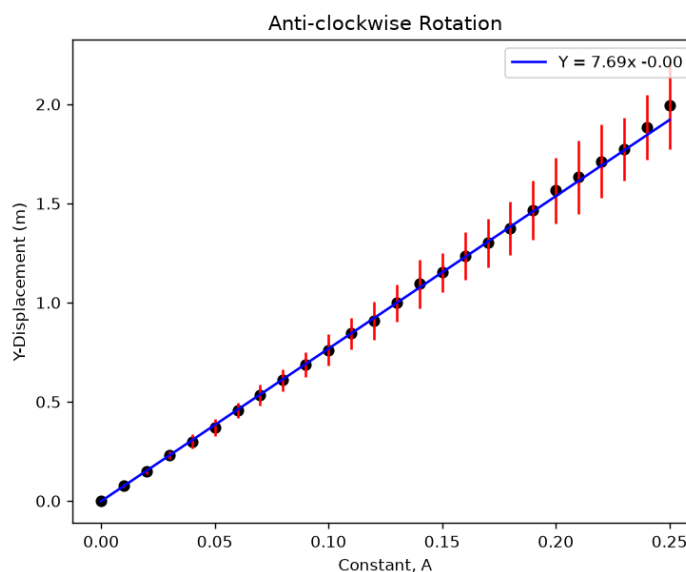


Figure 2: A graph showing the linear relationship between the constant A and the final y displacement of the curling stone, with standard deviation as error bars and a weighted line of best fit.

The line of best fit has a gradient of 7.69m and a y-intercept of 0, the line of best fit fits the data points well.

Conclusion:

Overall, the model produces a linear relationship between the constant A and y displacement value as expected, it also produces curl in the correct direction and of the predicted order of magnitude.

Future Improvements:

- Constant A should be a function of angular velocity and mass of the stone
- Not using a constant value for the variable t_b
- Accounting for pivoting motion
- Final curl should be independent of pebble density, the current model has a linear relationship between pebble density and final curl
- Using a distribution for pebble radius instead of using a constant value